

What challenges are faced when integrating 5G with IoT? Are they specific to India, or are they global?

Integrating IoT with 5G poses challenges globally, and are not limited to India. Legacy IoT devices, representing 87% of the current IoT landscape, are not continuously upgraded to keep pace with telecom advancements like 6G or 7G due to prohibitive costs. Thus, they require a bridging solution. The diversity of IoT devices from various manufacturers complicates integration efforts. Each brand typically operates within its own ecosystem, lacking compatibility with others. This fragmentation hinders connectivity across devices. Security concerns may arise as IoT devices communicate with networks, making it imperative to identify and track the source of data transmissions to prevent security breaches.

In India, railways and agriculture need real-time communication and data exchange among IoT devices. Upgrading existing IoT devices to 5G compatibility is impractical due to cost constraints, especially for large-scale deployments in rural areas. Cienra offers a cost-optimised solution, AADI, that assigns a unique ID to each IoT device to facilitate connectivity between legacy IoT devices and 5G networks.

How does your solution work?

We take legacy IoT connections—whether via Wi-Fi, LoRa, 4G, hotspot, or any other interface—and funnel them into our solution. After IP multiplexing and various operations, we bring it into our stack, where it follows a 3GPP protocol. Legacy IoT hardware primarily relies on SMS or minimal data transmission. AADI allows legacy IoT devices to connect, regardless of the network generation (G) involved, without using radio. Instead, we use layers unique to 5G and beyond,

approved by standards, allowing us to circumvent the need for radio transmission. Typically, data would pass through a base station antenna, but we bypass this step. Instead, we directly connect to the 5G public core or, alternatively, to our or any other 5G core, and then to the servers through broadband IP. These servers can be either at the edge or in the cloud—it is immaterial to us.

Are there any other companies offering similar solutions?

We have yet to encounter any direct competition in India, but we did meet some individuals in Barcelona this year who had similar solutions but no direct competition. We operate in a unique space. While big players like Cisco, Amazon, Google, and Microsoft dominate the cloud and edge computing realms, companies like Intel and Qualcomm lead the hardware sector. We occupy the middle ground, where there is less attention from major players focused on pushing their latest technologies. Even organisations like DCIS and government bodies have been collaborating with us in the telecom sector.

Who can benefit from such a solution?

There are three to four main customer segments, including integrators, who undertake projects to transform networks into 5G and cities into smart cities, which require solutions to connect legacy IoT devices. Another customer segment includes those involved in enabling government 5G deployments. We also work with telecom operators who seek solutions for specific use cases, such as converting factory flows into smart factories. Universities interested in setting up 5G labs are also part of our target audience, although their focus is more academic than commercial.

With so many target customers in the public sector, are there any regulatory compliances that need to be kept in mind?

There are specific requirements to consider for regulatory compliance, particularly in the telecom sector. For instance, certifications and approvals from organisations like CDOT are essential in India. Compliance with 3GPP specifications is mandatory; obtaining telecom certifications is important wherever the solution is deployed. Meeting these compliance standards also opens doors to recommendations from regulatory bodies and government agencies, establishing the solution's credibility and potential for widespread adoption.

What is the need for such a solution?

It is easier and more cost-effective to deploy this product than to change traditional IoT hardware solutions. For large-scale IoT deployments, such as those in smart cities, this approach reduces both upfront and ongoing costs, offering a more accessible pathway to implementation.

How scalable is this solution?

This solution is very scalable due to its cloud-based architecture, which makes it easy to deploy and maintain through cloud support and upgrades. Initial integration is a one-time task, and the product adheres to standard protocols, making scalability smooth and accessible.

Do you integrate with third-party technologies and systems as well?

Yes, we integrate with third-party technologies and systems. We are also exploring potential partnerships with various telecom operators. AADI is integrated with the 5G Core of many vendors as it complies with 3GPP 5G standards. **EFY**